ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

# FREQUENTIST VERSUS BAYESIAN ESTIMATION

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# Inferring an Unknown Parameter

We get data $x_1, \dots, x_n \sim p_\theta$ and would like to estimate $\theta$.

- **Frequentist: treat $\theta$ as an unknown, non-random quantity.**

  For instance, pick the maximum-likelihood value

$$\arg\max_\theta \Pr(x_1, \dots, x_n | \theta) = \arg\max_\theta \prod_{i=1}^{n} p_\theta(x_i).$$

- **Bayesian: treat $\theta$ as a random variable with a prior distribution $q_o$.**

  Given the data, the posterior distribution of $\theta$ is

$$q_n(\theta) = \Pr(\theta | x_1, \dots, x_n) \propto q_o(\theta) \Pr(x_1, \dots, x_n | \theta).$$

## Some Good References

- Andrew Gelman, John Carlin, Hal Stern, Donald Rubin. Bayesian Data Analysis.
- Kevin Murphy. Machine Learning: A Probabilistic Perspective.

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- Mathematical setup:
  - Let $\theta$ be the probability that a child is female.
  - Let $n$ be the number of children observed.

  Suppose we see $F$ females and $M$ males. Then $F \sim \text{binomial}(n, \theta)$.

- Bayesian inference: put a prior on $\theta$.
  Simple choice: uniform prior, $q_o(\theta) = 1$ for all $\theta \in [0.1]$.

- Laplace asked: What is the probability that $\theta < 0.5$?

- Uniform prior: $q_o \equiv 1$.
- What is the posterior $q_n$ after seeing $F = f, M = m$?